

Thesis presented to the Instituto Tecnológico de Aeronáutica, in partial fulfillment of the requirements for the degree of Doctor of Science in the Graduate Program of Engenharia Eletrônica e Computação, Field of Sistemas e Controle.

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**PREDICTIVE CONTROL OF SWITCHED AFFINE
SYSTEMS SUBJECTED TO DWELL TIME
CONSTRAINTS**

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1 Introduction

1.1 Motivation

The dynamics of switched affine systems can be described by a state equation of the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}, t \in \mathbb{R}_+$, where $x(t) \in \mathbb{R}^n$ is the state vector and $\sigma(t) \in \{1, 2, \dots, M\}$ is the switching state associated with the matrices $A_{\sigma(t)} \in \mathbb{R}^{n \times n}$ and the vectors $b_{\sigma(t)} \in \mathbb{R}^n$ at each time t .

There is extensive research in the synthesis of controllers for switched affine systems. However, the design procedures are typically concerned with driving the state $x(t)$ to the vicinity of a given target, without considering the subsequent evolution of the state in that region. This issue is especially relevant when the switching state $\sigma(t)$ is subjected to dwell-time constraints¹; in that case, rather than driving the state to an equilibrium point, the design should be aimed at achieving a suitable limit cycle.

O que esses tres trabalhos nao fizeram, no que diz respeito ao escopo desse doutorado? ou seja, o que estamos fazendo que eles nao fizeram? eh o tratamento de restricoes de dwell-time?

A resposta para essa pergunta servira de base para a ligacao do texto com o proximo paragrafo.

- the work treats the case of changing dynamics
- computes the dwell-time constraints
- has the convenience of solving the problem with quadratic programming convex optimization

The present work extends the problem to work with changing dynamics at switching times and applies the dwell-time constraints using predictive control. The formulation of the solution also has the convenience of solving the problem with quadratic programming convex optimization.

This problem has been addressed, for instance, by Benmiloud *et al.* (2019) who uses a hybrid Poincaré map approach to stabilize a switched affine system into the desired

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 , i.e. constraints on the minimum time between successive changes in $\text{state}(t)$.

limit cycle and a DC-DC converter was employed as an illustrative example. Egidio *et al.* (2020b) proposed the control design of a state-dependent switching function for discrete-time switched affine systems to stabilize a limit cycle determined according to criteria of interest related to the steady-state behavior. The control design applies a switched rule to orchestrate the state trajectories based on a time-varying convex Lyapunov function that drives the system from the initial condition toward the desired limit cycle. The proposed method uses a voltage regulation of a simulated DC-DC multicellular converter as an illustrative example. This work was extended to continuous time switched-affine systems by Egidio *et al.* (2020a)

Some studies form the basis for the present work:

Vieira (2013) solved the problem for a particular class of affine systems. Here the controller drove the system to a target state using MILP (Mixed-Integer Linear Programming) and the cyclic trajectory was a consequence of the states as the system approached the target.

Patino *et al.* (2010) defined the target trajectory using nonlinear programming and used this target trajectory in the predictive control in terms of errors.

Marcolino (2018) proposed a solution with a controller law that drives the switched affine system with dwell-time constraints to a target cyclic trajectory. In this work, the problem is addressed in two stages: first, the problem is interpreted as a convex optimization where the author can find a targeted cyclic trajectory that minimizes a cost function associated with the system performance; then a second part that uses the target trajectory as input and computes perturbation at the switching instants of the reference control signal using predictive control.

1.2 Contributions of the present work

This present work proposes to expand the solution from Marcolino (2018) and provide a broader solution that can be used for systems that have their dynamics changing over time.

In a system that can be represented by a switched affine system with the form $\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)}$ there are cases in which the dynamic response of the system can change depending on the operation mode. This can be represented in switched affine systems with proper matrices $A_{\sigma(t)}$ and $b_{\sigma(t)}$ for each operation mode. With the current implementation of Marcolino (2018) considering an invariant system during its operation, there is a need for expanding the solution so it can represent more systems and solve more engineering problems that can be expressed as switched affine systems.

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Mencionar que estes são trabalhos que usaram controle preditivo para levar em conta restrições de dwell time em sistemas lineares com atuadores chaveados, que são um caso particular de switched affine systems.
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-  Number: 3 Author: Kawakami Subject: Sticky Note Date: 26/12/2023 11:22:54
Citar o artigo de Marcolino, em lugar da tese.
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to general case of switched affine systems.
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an extension of
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switched affine systems
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- ☐ Number: 7 Author: Kawakami Subject: Rectangle Date: 26/12/2023 11:25:15
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-  Number: 8 Author: Kawakami Subject: Sticky Note Date: 26/12/2023 11:26:10
Este trecho não será necessário. No parágrafo anterior, já será explicado que o trabalho de Marcolino envolvia uma classe particular de switched affine systems.
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-  Number: 9 Author: Kawakami Subject: Sticky Note Date: 26/12/2023 11:33:23
Aqui seria importante explicar os três elementos básicos da abordagem de Marcolino, que serão também adotados aqui: (1) Determinação de uma trajetória alvo periódica, (2) linearização da cycle-to-cycle dynamics do sistema em torno da trajetória alvo, considerando perturbações nos instantes de chaveamento como variáveis de controle e (3) uso do modelo linearizado em conjunto com uma formulação de controle preditivo com imposição de restrições de dwell time.

2 Determination of Target Periodic Trajectories

The goal is to drive a system defined by an affine dynamic equation (eq. 2.1) to an optimal periodic trajectory, so the first step is to generate this trajectory. This trajectory is submitted to restrictions imposed by the dynamic of the system and also switching time restrictions.

$$\dot{x}(t) = A_{\sigma(t)}x(t) + b_{\sigma(t)} \quad (2.1)$$

Following the approach from Patino *et al.* (2010) we must establish a trajectory that yields the optimal time sequence $\mathbb{J}^\infty$, the modes sequence Ω^∞ , and its length $s^\infty (1 < s^\infty < s_{max})$. To reduce the oscillations around a desired operation point, we choose an integral quadratic criterion centered on the average reference value $\bar{x}^\infty$:

$$J(\tau^\infty, \Omega^\infty, s^\infty) = \min_{\tau, \Omega, s} \int_0^{T_p} [x(t) - \bar{x}^\infty]^T Q [x(t) - \bar{x}^\infty] dt \quad (2.2)$$

where $Q = Q^T > 0$ and T_p is the period of the cycle.

Since the trajectory is cyclic we can impose the constraint:

$$x(0) = x(T_p) \quad (2.3)$$

2.1 Optimization

2.1.1 System Dynamics and Sequence

To begin, we create a list where each of its elements signifies the dynamics of the system. This is exemplified by the matrices A, B, C, and D. The order in which the elements appear in this list is defined by the sequence of modes. In simpler terms, the positioning of these elements signifies the order they will be executed.

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Idealmente, cada equação deve ser apresentada antes de ser referenciada no texto. Neste caso, por exemplo, o melhor seria começar o parágrafo mais ou menos da seguinte forma: "Consider a switching affine system of the form

$\dot{x}(t) = \dots$ (2.1)

The goal is ..." Idem no restante do texto. Vale notar, contudo, que trechos como o da equação (2.3) nesta página estão OK, pois não está sendo usada uma referência de numeração antes da equação.

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Definir.

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Este trecho já deveria fazer parte da seção 2.1.

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Seria conveniente também incluir $x(0)$ com variável de otimização, pois a trajetória $x(\cdot)$ depende de $x(0)$.

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Não é mais assim. Vide a Figura 2.1.

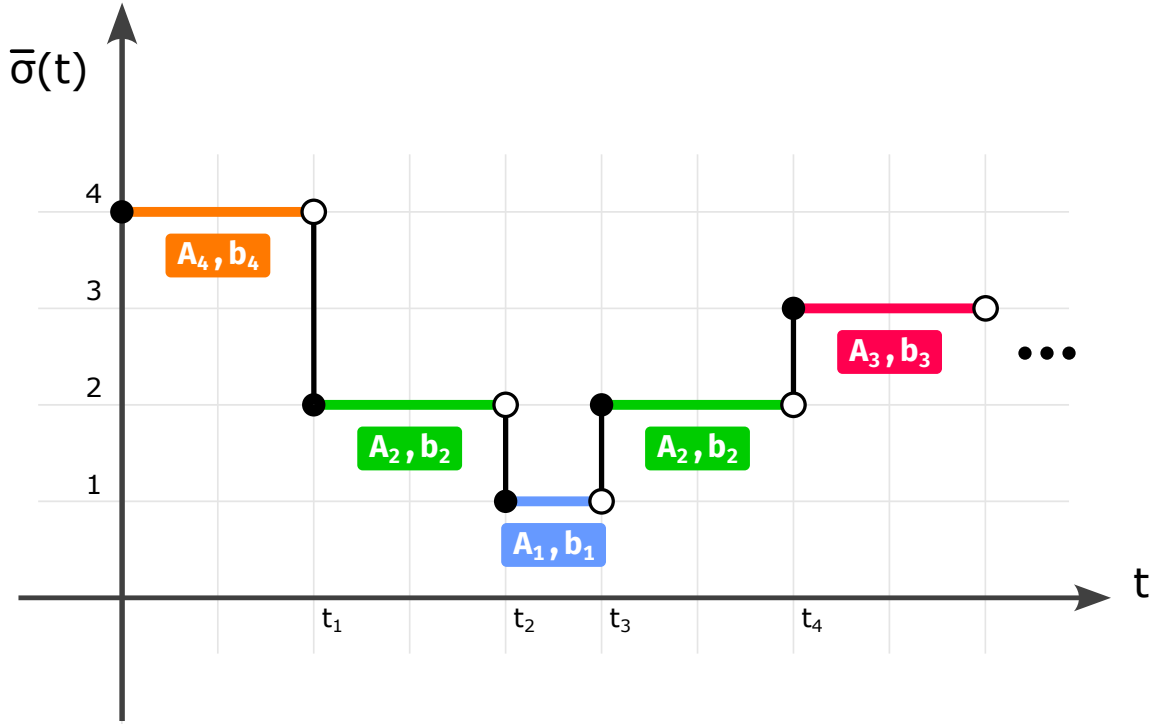


FIGURE 2.1 – Change in dynamics over trajectory time

¹ Figure 2.1, we see a trajectory example. For this illustration, the system's list of dynamics is $\{[A_1, b_1], [A_2, b_2], [A_3, b_3], [A_4, b_4]\}$, and the sequence of modes is $[4, 2, 1, 2, 3]$. It's worth noting that a particular system dynamic can be employed multiple times or might not be used at all, depending on the desired target trajectory. The sequence of modes and the maximum period for the cycle are predetermined for this problem and the goal is to achieve the best trajectory that minimizes the cost function.

2.1.2 Constraints Definition ²

³ Once the list and sequence have been established, we need to apply the constraints. These constraints dictate the cycle's maximum duration, as well as the minimum and the ⁴maximum time permitted between switches in modes.

2.1.3 Optimization with FMINCON ⁵

For optimization purposes, the function FMINCON is employed. This function works in conjunction with a cost function. The cost function is designed to accept switching times generated from the optimization process. Its primary goal is to evaluate and discern the most cost-effective result that respects the imposed constraints.

FMINCON is a method of optimization in MATLAB's libraries that implements the

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Este esclarecimento precisa ser apresentado anteriormente, quando estiverem sendo definidos τ_{inf} e Ω_{inf} .
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-  Number: 3 Author: Kawakami Subject: Sticky Note Date: 26/12/2023 11:46:38
As restrições precisam ser apresentadas matematicamente, usando a notação apresentada na definição de τ_{inf} .
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Acho que não precisaríamos impor restrições deste tipo, já que há uma restrição sobre a duração máxima do ciclo.
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Esta subseção 2.1.3 precisaria vir depois do desenvolvimento apresentado na Seção 2.2, que ainda faz parte da formulação do problema de otimização em si.

method SQP (Sequential Quadratic Programming). The basic SQP algorithm is described in Chapter 18 of Nocedal e Wright (2006).

2.1.4 Cost Function and Trajectory Simulation

Taking a closer look at the cost function, its role is to simulate the trajectory using the dynamics taken from the previously defined list, all in line with the mode sequence. This trajectory is cyclic and to enforce this cyclic nature, the initial state's condition is calculated in such a way that the end product is a cyclic trajectory.



2.1.5 Cost Computation

The method of determining the cost is by computing the integral of the trajectory, centered around the average reference value, denoted as $\bar{x}^\infty$. The result is compliant with the system's dynamic constraints and also matches the properties of the final trajectory.

2.2 Calculating x_0

We need to compute the initial condition for the system for each evaluation of the cost function so that we have a periodic trajectory, meaning that the last states of the trajectory we the same as the initial condition.

We can have different dynamics for each switching instant, so the system behaves as

$$\begin{aligned}
 [0 \rightarrow t_1] : \dot{x} &= A_{\sigma(0)}x + b_{\sigma(0)} \\
 [t_1 \rightarrow t_2] : \dot{x} &= A_{\sigma(t_1)}x + b_{\sigma(t_1)} \\
 &\vdots \\
 [t_{N-1} \rightarrow t_N] : \dot{x} &= A_{\sigma(t_{N-1})}x + b_{\sigma(t_{N-1})}
 \end{aligned} \tag{2.4}$$

We can rewrite the solution for the first region as

$$\begin{aligned}
 x(t_1) &= e^{A_{\sigma(0)}t_1}x(0) + \int_0^{t_1} e^{A_{\sigma(0)}(t_1-\tau)}b_{\sigma(0)}d\tau \\
 x(t_1) &= e^{A_{\sigma(0)}t_1}x(0) + \int_0^{t_1} e^{A_{\sigma(0)}(t_1-\tau)}b_{\sigma(0)}d\tau \\
 x(t_1) &= F_0x(0) + g_0
 \end{aligned} \tag{2.5}$$

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Estas duas subsecções podem ser juntadas em um único trecho, a ser incluído logo antes de se falar do fmincon.

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